

Xuenan Li

Department of Applied Physics and Applied Mathematics
Columbia University
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Research Interests

Calculus of Variations, Partial Differential Equations, and Spectral Theory
with particular interests in systems with multi-scale

Rigidity Theory
with particular interests in bar frameworks

Academic Position

2023- Chu Assistant Professor in Applied Mathematics
COLUMBIA UNIVERSITY

Education

2018- 2023 Ph.D. in Mathematics
COURANT INSTITUTE OF MATHEMATICAL SCIENCES, NEW YORK UNIVERSITY, NY, USA
Advisor: Robert V. Kohn
Thesis: Guest-Hutchinson modes and the macroscopic energy of the Kagome lattice and related systems

2015-2018 B.S. in Mathematics
UNIVERSITY OF MICHIGAN, ANN ARBOR, MI, USA

Publications & Preprints

Submitted & in preparation:

Li, Xuenan and Kohn, Robert V. , *Some results on the Guest-Hutchinson modes and periodic mechanisms of the Kagome lattice*, J Mech Phys Solids 178 (Sept 2023) 105311, [published version](#), [arXiv version](#).

Li, Xuenan, Holmes-Cerfon, Miranda and Santangelo, Christian D. (2025) *A Constrained Saddle Search Approach for Constructing Singular and Flexible Bar Frameworks*, submitted, [arXiv version](#).

Li, Xuenan and Kohn, Robert V. (2025) *The effective energy of a lattice system*, [arXiv version](#).

Li, Xuenan, Holmes-Cerfon, Miranda and Santangelo, Christian D. (2025) *A Constrained Optimization Approach for Constructing Rigid Bar Frameworks with Higher-order Rigidity*, in preparation.

Li, Xuenan and Kohn, Robert V. (2025) *A nonlinear homogenization-based perspective on the soft modes and effective energy of conformal lattice systems*, in preparation.

Li, Xuenan and Weinstein, Michael I. (2025) *The discrete domain wall in the slowly-perturbed SSH model*, in preparation.

Teaching Experiences

Spring 2025	<i>Instructor</i> APAM E4101 Dynamical systems Columbia University
Fall 2024	<i>Instructor</i> APAM E4204 Functions of a complex variable Columbia University
Fall 2023	<i>Instructor</i> APAM E4204 Functions of a complex variable Columbia University
Spring 2024	<i>Instructor</i> APAM E4101 Dynamical systems Columbia University
Fall 2024	<i>Instructor</i> APAM E4204 Functions of a complex variable Columbia University
Fall 2022	<i>Teaching Assistant</i> MATH-UA.0262-007 Ordinary Differential Equations Courant Institute, New York University
Spring 2022	<i>Teaching Assistant</i> MATH-UA.0140-007: Linear Algebra Courant Institute, New York University
Fall 2021	<i>Teaching Assistant</i> MATH-UA.0263-002: Partial Differential Equations Courant Institute, New York University
Spring 2021	<i>Teaching Assistant</i> MATH-UA.0121-017: Calculus I Courant Institute, New York University

Academic Services

Spring 2025	Thesis defense committee member: PhD Candidate: <i>Edith Jin Zhang</i> , Department of Applied Physics and Applied Mathematics Dissertation Title: <i>Ginzburg–Landau Functionals and Reaction–diffusion Equations in the Large-Graph Limit</i> .
Fall 2024 - now	Master Student Research Program at Columbia University with <i>Wendi Lu</i> ;
Fall 2024	Thesis defense committee member: PhD Candidate: <i>Zirui Xu</i> , Department of Applied Physics and Applied Mathematics Dissertation Title: <i>Ohta–Kawasaki Energy and Its Phase-Field Simulation</i> .
Summer 2024	Undergraduate Student Research Program at Columbia University with two undergraduate students <i>Mihnea Leonte</i> and <i>Andrew Yang</i> ;

Program title: *Designing singular structures in bar-joint networks*;
Funded by *Bonomi Scholars* at Columbia University.

Update (Spring 2025): Mihnea received the Applied Mathematics Faculty Award; Andrew received the Applied Physics Faculty Award.

Honors & Awards

2025	AWM Travel Grant
2023	SIAM Student Travel Award, 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023)
2022	Sandra Bleistein Prize, New York University
2022	SIAM Student Travel Award, SIAM Annual Meeting (AN22)
2018–2023	New York University MacCracken Graduate Scholarship, New York University

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<https://xuenanli.github.io/>